Prime-Square Tail Rate and a Quadratic Memory Remainder for Phasewise Chowla-Free Memory

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August 7, 2026

Abstract

Within the fixed finite-clock, universally distance-two-safe phasewise lag-two class whose interpolation coefficient $c_{11}(r)$ vanishes at every phase, RH-379 identifies an all-clock supremum B_∞ and RH-380 proves that the square-clock values $G(q_y)$ increase strictly to it. We determine the first-order rate of that convergence. Put

$$a_{j+1} = \frac{1}{p_{j+1}^2 - 1}, \qquad T_y = \sum_{j \geq y} a_{j+1},$$

where $3 = p_1 < p_2 < \cdots$ are the odd primes, and let

$$e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2}\right), \qquad X_\infty = \frac{2e_4 - 4e_5 + 6e_6 - 8e_7 + 10e_8}{e_1}.$$

The run formula gives $X_\infty \geq 6e_8/e_1 > 0$. A factorwise Euler-tail comparison proves $|X_j - X_\infty| \leq 170 T_j$. The exact RH-379 product for H_{j+1} gives

$$0 \leq \frac{4}{\pi^2} - H_{j+1} \leq \frac{4}{\pi^2} T_{j+1},$$

and the normalized memory statistic satisfies $0 \leq M_j/A_j \leq 1$. Telescoping the exact RH-380 increment and using two exact tail-sum identities yields, for every $y \geq 1$,

$$\left| B_\infty - G(q_y) - \frac{2X_\infty}{\pi^2} T_y \right| \leq \frac{342}{\pi^2} T_y^2.$$

Consequently

$$\frac{B_\infty - G(q_y)}{T_y} \longrightarrow \frac{2X_\infty}{\pi^2} > 0.$$

No prime number theorem is used. The result supplies neither a second-order coefficient nor a p_y-scale asymptotic, growing-clock theorem, adaptive capacity limit, operator, trace formula, zero identification, or implication for the Riemann hypothesis.

Keywords: Möbius function; prime-square Euler tail; square clocks; quantitative convergence; lag-two memory.

1 Frozen class and exact predecessor inputs

The theorem concerns exactly the class isolated in RH-379 and retained in RH-380 [1, 2]. We begin by fixing its quantifiers.

Definition 1.1 (Fixed-clock phasewise class)**.** Fix an integer $q \geq 1$ before taking $N \to \infty$. For every $r \in \mathbb{Z}/q\mathbb{Z}$, choose a table

$$f_r : \{-1, 0, 1\}^2 \longrightarrow \{-1, +1\}, \qquad \epsilon_n = f_{n \bmod q}\big(\mu(n-2), \mu(n)\big).$$

The family is universally distance-two-safe if no ternary input can give $\epsilon_n = \epsilon_{n+2} = +1$. It is in the present class only when the coefficient $c_{11}(r)$ of $\mu(n-2)\mu(n)$ in the RH-379 interpolation is zero for every phase. The exact fixed-clock optimum is denoted by $G(q)$.

The phasewise condition is not cosmetic. If $c_{11}(r) \neq 0$, the limiting expansion contains a phase-weighted shift-two Möbius correlation that is not controlled by the frozen sources. Nothing below averages or cancels those missing terms.

Let $3 = p_1 < p_2 < \cdots$ be the odd primes and define the square-clock data

$$P_y = \prod_{i \le y} p_i^2, \qquad q_y = 4P_y, \qquad A_y = \prod_{i \le y} (p_i^2 - 1), \qquad D_y = \prod_{i \le y} (p_i^2 - 2). \tag{1}$$

RH-374 associates a cyclic odd squarefree-support word to P_y. Its positive runs have lengths at most eight; write $R_\ell^{(y)}$ for the number of runs of length ℓ [3]. With

$$E_m^{(y)} = \prod_{i \le y} \left(1 - \frac{m}{p_i^2}\right), \tag{2}$$

the exact run formula is

$$R_\ell^{(y)} = \begin{cases} P_y(E_\ell^{(y)} - 2E_{\ell+1}^{(y)} + E_{\ell+2}^{(y)}), & 1 \le \ell \le 7, \\ P_y E_8^{(y)}, & \ell = 8. \end{cases} \tag{3}$$

Set

$$\mathcal{E}_y = R_2^{(y)} + R_4^{(y)} + R_6^{(y)} + R_8^{(y)}, \qquad L_y = 2R_2^{(y)} + 4R_4^{(y)} + 6R_6^{(y)} + 8R_8^{(y)}, \tag{4}$$

$$M_y = 2R_3^{(y)} + 4R_5^{(y)} + 6R_7^{(y)}. \tag{5}$$

RH-380 proves the exact increment

$$\begin{aligned} G(q_{j+1}) - G(q_j) = {} & \frac{2(L_j - 2\mathcal{E}_j)}{\pi^2 A_j (p_{j+1}^2 - 1)} \\ & + \frac{M_j}{A_j(p_{j+1}^2 - 1)} \left(\frac{4}{\pi^2} - H_{j+1}\right), \end{aligned} \tag{6}$$

where $H_j = \kappa_2 A_j / D_j$, and it records

$$G(q_j) \longrightarrow B_\infty. \tag{7}$$

The limit in (7) is the RH-379 all-clock supremum. It is a cofinal limit after the fixed-clock N-limits, not a growing choice $q = q(N)$.

2 Prime-square tails and the normalized run statistic

For $j \ge 1$, put

$$a_{j+1} = \frac{1}{p_{j+1}^2 - 1}, \qquad T_j = \sum_{k \ge j} a_{k+1}. \tag{8}$$

The notation deliberately keeps the successor index visible: the increment from q_j to q_{j+1} carries a_{j+1}.

For $1 \le m \le 8$, define

$$e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2}\right), \qquad U_m^{(j)} = \frac{E_m^{(j)}}{E_1^{(j)}}, \qquad u_m = \frac{e_m}{e_1}. \tag{9}$$

All factors in these products are positive. Their convergence and positivity follow elementarily from convergence of $\sum_{n \ge 3} n^{-2}$. The RH-374 normalization is $e_1 = 8/\pi^2$ [3].

Definition 2.1 (Normalized deterministic numerator)**.** Define

$$X_j = \frac{L_j - 2\mathcal{E}_j}{A_j}, \qquad X_\infty = 2u_4 - 4u_5 + 6u_6 - 8u_7 + 10u_8. \tag{10}$$

Equivalently,

$$X_\infty = \frac{2e_4 - 4e_5 + 6e_6 - 8e_7 + 10e_8}{e_1}.$$

Lemma 2.2 (Exact finite Euler form)**.** *For every $j \geq 1$,*

$$X_j = 2U_4^{(j)} - 4U_5^{(j)} + 6U_6^{(j)} - 8U_7^{(j)} + 10U_8^{(j)}. \tag{11}$$

Moreover,

$$X_\infty \geq \frac{6e_8}{e_1} > 0. \tag{12}$$

Proof. The run identity in RH-380 is

$$L_j - 2\mathcal{E}_j = 2R_4^{(j)} + 4R_6^{(j)} + 6R_8^{(j)}.$$

Since $A_j = P_j E_1^{(j)}$, substituting (3) gives

$$\begin{aligned} X_j &= 2\frac{E_4^{(j)} - 2E_5^{(j)} + E_6^{(j)}}{E_1^{(j)}} + 4\frac{E_6^{(j)} - 2E_7^{(j)} + E_8^{(j)}}{E_1^{(j)}} + 6\frac{E_8^{(j)}}{E_1^{(j)}} \\ &= 2U_4^{(j)} - 4U_5^{(j)} + 6U_6^{(j)} - 8U_7^{(j)} + 10U_8^{(j)}. \end{aligned}$$

Passing to convergent Euler products proves the formula for X_∞. The two finite second differences in the first line are normalized run counts and hence nonnegative. Their limits remain nonnegative. Dropping them leaves $6e_8/e_1$, which is strictly positive. □

Lemma 2.3 (Factorwise tail bound)**.** *For every $j \geq 1$,*

$$\boxed{|X_j - X_\infty| \leq 170T_j.} \tag{13}$$

Proof. For $m \in \{4, 5, 6, 7, 8\}$, division of the Euler factors gives the exact tail identity

$$\frac{u_m}{U_m^{(j)}} = \prod_{p > p_j} \frac{1 - m/p^2}{1 - 1/p^2} = \prod_{p > p_j} \left(1 - \frac{m-1}{p^2 - 1}\right). \tag{14}$$

For numbers $0 \leq v_i \leq 1$, the elementary product union bound $0 \leq 1 - \prod_i (1 - v_i) \leq \sum_i v_i$ follows first for finite products and then by monotone passage to the limit. Also $0 < U_m^{(j)} \leq 1$. Therefore

$$0 \leq U_m^{(j)} - u_m \leq (m-1)T_j. \tag{15}$$

Apply the triangle inequality to (11). The coefficient ledger is

$$2 \cdot 3 + 4 \cdot 4 + 6 \cdot 5 + 8 \cdot 6 + 10 \cdot 7 = 6 + 16 + 30 + 48 + 70 = 170,$$

which proves (13). □

3 The H tail and the memory statistic

Lemma 3.1 (Exact H product and its tail loss)**.** *For every $j \geq 1$,*

$$\frac{H_{j+1}}{4/\pi^2} = \prod_{p>p_{j+1}} \left(1 - \frac{1}{p^2-1}\right), \tag{16}$$

and hence

$$0 \leq \frac{4}{\pi^2} - H_{j+1} \leq \frac{4}{\pi^2} T_{j+1}. \tag{17}$$

Proof. The exact RH-379 product is $H_j = \kappa_2 A_j / D_j$, where $\kappa_2 = \prod_p (1 - 2/p^2)$. Separating the prime 2 and cancelling the finite odd-prime factors gives

$$H_{j+1} = \frac{1}{2} E_1^{(j+1)} \prod_{p>p_{j+1}} \left(1 - \frac{2}{p^2}\right).$$

Because $4/\pi^2 = e_1/2$, division yields

$$\frac{H_{j+1}}{4/\pi^2} = \prod_{p>p_{j+1}} \frac{1-2/p^2}{1-1/p^2} = \prod_{p>p_{j+1}} \left(1 - \frac{1}{p^2-1}\right).$$

The same product union bound used in Lemma 2.3 now gives (17). ☐

Lemma 3.2 (Quadratic-memory normalization)**.** *For every $j \geq 1$,*

$$0 \leq \frac{M_j}{A_j} \leq 1. \tag{18}$$

Proof. Nonnegativity is immediate from (5). An odd run of length ℓ contributes $\ell - 1 \leq \ell$ to M_j. Thus M_j is at most the number of positive sites lying in odd runs, which is at most the total number of positive sites. The latter is exactly $P_j E_1^{(j)} = A_j$. This proves (18). ☐

The word "quadratic" in the title refers to the contribution of this memory statistic to the final T_y^2 remainder. It does not announce a quadratic coefficient expansion.

4 Two exact tail-sum identities

Lemma 4.1 (Tail algebra)**.** *For every $y \geq 1$,*

$$\sum_{j \geq y} a_{j+1} T_j = \frac{1}{2}\left(T_y^2 + \sum_{j \geq y} a_{j+1}^2\right) \leq T_y^2, \tag{19}$$

$$\sum_{j \geq y} a_{j+1} T_{j+1} = \frac{1}{2}\left(T_y^2 - \sum_{j \geq y} a_{j+1}^2\right) \leq \frac{1}{2} T_y^2. \tag{20}$$

Proof. Since $T_{j+1} = T_j - a_{j+1}$,

$$a_{j+1} T_j = \frac{1}{2}(T_j^2 - T_{j+1}^2 + a_{j+1}^2),$$
$$a_{j+1} T_{j+1} = \frac{1}{2}(T_j^2 - T_{j+1}^2 - a_{j+1}^2).$$

Sum these identities from $j = y$ to a finite endpoint and then let that endpoint tend to infinity. The tails tend to zero because

$$0 < T_y \leq \sum_{n>p_y} \frac{1}{n^2-1} = \frac{1}{2}\left(\frac{1}{p_y} + \frac{1}{p_y+1}\right) \longrightarrow 0. \tag{21}$$

Here positivity holds for every finite y because there are infinitely many odd primes. Finally, $\sum a_{j+1}^2 \leq (\sum a_{j+1})^2 = T_y^2$, which gives both displayed inequalities. No estimate for the distribution of primes is involved. ☐

5 First-order gap rate

Proposition 5.1 (Exact infinite increment sum)**.** *For every $y \geq 1$,*

$$B_\infty - G(q_y) = \sum_{j\geq y}\Bigg[\frac{2X_j}{\pi^2}a_{j+1} + \frac{M_j}{A_j}a_{j+1}\left(\frac{4}{\pi^2} - H_{j+1}\right)\Bigg]. \tag{22}$$

Proof. For a finite $K > y$, telescope the exact RH-380 increment (6) from $j = y$ through $K - 1$ and use $X_j = (L_j - 2\mathcal{E}_j)/A_j$ and $a_{j+1} = (p_{j+1}^2 - 1)^{-1}$. Then let $K \to \infty$ and invoke the already proved cofinal limit (7). This is a limit of finite telescoping identities after all fixed-clock N-limits have been taken; there is no exchange of an N-limit with a j-limit. □

Theorem 5.2 (Prime-square tail rate)**.** *For every $y \geq 1$,*

$$\boxed{\left|B_\infty - G(q_y) - \frac{2X_\infty}{\pi^2}T_y\right| \leq \frac{342}{\pi^2}T_y^2.} \tag{23}$$

Consequently,

$$\boxed{\frac{B_\infty - G(q_y)}{T_y} \longrightarrow \frac{2X_\infty}{\pi^2} > 0.} \tag{24}$$

Proof. Subtract $(2X_\infty/\pi^2)T_y$ from (22). Since $T_y = \sum_{j\geq y} a_{j+1}$, the first remainder is bounded using Lemma 2.3 and (19):

$$\left|\sum_{j\geq y}\frac{2(X_j - X_\infty)}{\pi^2}a_{j+1}\right| \leq \frac{340}{\pi^2}\sum_{j\geq y}a_{j+1}T_j \leq \frac{340}{\pi^2}T_y^2. \tag{25}$$

The memory summand is nonnegative. Lemmas 3.1 and 3.2, followed by (20), give

$$0 \leq \sum_{j\geq y}\frac{M_j}{A_j}a_{j+1}\left(\frac{4}{\pi^2} - H_{j+1}\right) \tag{26}$$

$$\leq \frac{4}{\pi^2}\sum_{j\geq y}a_{j+1}T_{j+1} \leq \frac{2}{\pi^2}T_y^2. \tag{27}$$

Adding (25) and (27) proves (23); the ledger is $340 + 2 = 342$.

For each finite y, $T_y > 0$ by (21). Divide (23) by T_y, then use $T_y \to 0$ from the same elementary integer-square comparison. Equation (24) follows, and its limit is positive by (12). □

Remark 5.3 (No prime number theorem)*.* The scale in Theorem 5.2 is the intrinsic prime-square tail T_y. We never replace it by a function of p_y. The only tail comparison is (21), obtained by telescoping the integer series $(n^2 - 1)^{-1}$. Thus the proof uses neither the prime number theorem nor a weaker prime-counting asymptotic.

6 Exact artifact and finite diagnostics

The standard-library artifact has two distinct layers. First, exact `Fraction` arithmetic regenerates the run statistics, normalized Euler ratios, RH-380 increments, and finite versions of both identities in Lemma 4.1. The independently frozen six-row fixture has canonical SHA-256

`d55fd48071eb5b88c054f3d34329f274f792f2bbd859b4ab98e31b5b7020beb8`.

Second, directed decimal arithmetic at precision 60 is converted to exact rational endpoints. It enumerates the 9592 primes through 100000 and bounds every omitted prime by the exact integer tail

$$\sum_{n>100000} \frac{1}{n^2 - 1} = \frac{200001}{20000200000}. \tag{28}$$

All comparisons fail closed if outward upper and lower endpoints overlap in the wrong order. The rows $y = 1, 2, 3, 5, 10, 25$ reproduce the 170 and 342 bounds; they do not prove the all-y theorem by fitting.

For orientation only, the directed product layer encloses

$$0.46967971561917 < X_\infty < 0.46993774873491.$$

These decimals are a finite diagnostic. Positivity in the theorem is the exact run argument (12), not this numerical enclosure.

The release builder locks exactly 25 immutable inputs: seven RH-374 files, eight RH-379 files, eight RH-380 files, and two RH-MVP2 archive summaries. It checks both the live SHA-256 and byte identity with each declared release commit. Mutable project instructions and handoff state are deliberately excluded. The result ledger has a recursively closed Draft 2020-12 schema, and archive replay rejects missing members, unsafe paths, duplicate JSON keys, hash drift, release-commit rebinding, or a semantic PDF that differs from `main.pdf`.

Reproduction from this directory is

```
make result
make test
make pdf
make archive
```

7 Scope, limitations, and next edge

The theorem closes one precise question: it identifies the first-order prime-square-tail scale of the square-clock nonattainment gap and proves an explicit quadratic remainder inside the frozen RH-379 class. Its limits are equally precise.

- The theorem is phasewise $c_{11} = 0$. It does not cover unrestricted lag-two tables or assume the missing phase-weighted shift-two correlation cancellation.
- The constant 342 is explicit and uniform for $y \geq 1$, but no optimality is claimed for either 170 or 342.
- No exact second-order coefficient is identified. In particular, the theorem does not collapse all quadratic information to one multiple of T_y^2.
- No asymptotic for T_y in terms of p_y, no growing clock $q(N)$, and no adaptive-capacity convergence is proved.
- Finite product rows and decimal enclosures are reproduction only; they are not finite-order regression promoted to an all-order claim.
- No canonical intrinsic dynamical spectral determinant, time-oriented scattering completion, self-adjoint generator, von Mangoldt prime-power trace identity, or completed-zeta divisor equality is constructed. Gates A–E remain false/open [4].
- There is no Hilbert–Pólya operator, no identification of Riemann zeros, and no implication for the Riemann hypothesis.

The immediate within-class next edge is a genuinely second-order analysis of the Euler tails and of M_j/A_j. Such a successor must retain every independent quadratic tail scale that survives the exact algebra and prove an all-order remainder; this paper does not state its coefficient. The first class-enlargement trigger remains a theorem for the required phase-weighted shift-two Möbius correlations.

Declarations

Data and code availability. All theorem inputs, exact source code, tests, closed result schema, source locks, and archive manifests are contained in the repository paper directory. No private dataset is used.

Author contributions. The RH research program performed conceptualization, formal analysis, software construction, verification, artifact curation, adversarial review, and manuscript preparation.

Funding. No external funding is declared.

Competing interests. No competing interests are declared.

Ethics and human participants. This mathematical and computational study uses no human participants, animals, or personal data; ethics approval and consent are not applicable.

AI assistance disclosure. AI-assisted tools supported source triage, symbolic checking, deterministic artifact generation, drafting, and adversarial review. Every released claim is constrained by immutable repository sources and reproducible verification artifacts; responsibility for the scoped mathematical statements remains with the research program.

References

[1] RH research program. Phasewise chowla-free lag-two memory: Exact max-plus optimization and the all-clock supremum, 2026. Frozen repository release RH-379.

[2] RH research program. Square-clock monotonicity and finite-clock nonattainment for phasewise chowla-free memory, 2026. Frozen repository release RH-380.

[3] RH research program. Square clocks, run recurrences, and an euler-product floor for mobius distance-two capacity, 2026. Frozen repository release RH-374.

[4] Liang Wang. Conditional prime dynamics: a corpus frontier synthesis—provenance, branch separation, and the unpaid theorem budget in rh-1–rh-361, 2026. Frozen repository release RH-MVP2, commit c0aed13a34b8bbc53061aed23738660adcd3624c.